$$R(w)=csc(7w)$$
$$R(w)=\frac{1}{\sin(7w)}$$

$$\frac{dR}{dw}=\frac{dR}{dcsc}\cdot\frac{dcsc}{Dw}$$
$$\frac{dR}{dw}=-7\,csc(7w)\cot(7w)$$